MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 8919 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Passive TR-limiter, bandpass, frequency range 5,395 to 5,905 MHz.

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	tp	TA	Altitude
Unit:	kw	---	µs	°C	feet
Maximum:	300	0.001	1.15	71	10,000
Minimum:	4	---	---	10	---

PHYSICAL CHARACTERISTICS: See figure 1.

Mounting: Mates with UG-406B/U choke flange or equivalent

Weight: 2 pounds (approximate)

TEST CONDITIONS:

Parameter:	Incident power	tp	prr	F
Unit:	kw	µs	pps	MHz
Tolerance:	±10%	±0.10	---	±5%
Test condition 1:	40	1.0	1,000	F3
Test condition 2:	70	1.0	1,000	F3
Test condition 3:	300	1.0	1,000	F3

Frequency		
F	MHz	±
1	5,395	0.1%
2	5,450	0.1%
3	5,650	0.1%
4	5,825	0.1%
5	5,905	0.1%

GENERAL:

Qualification: Required

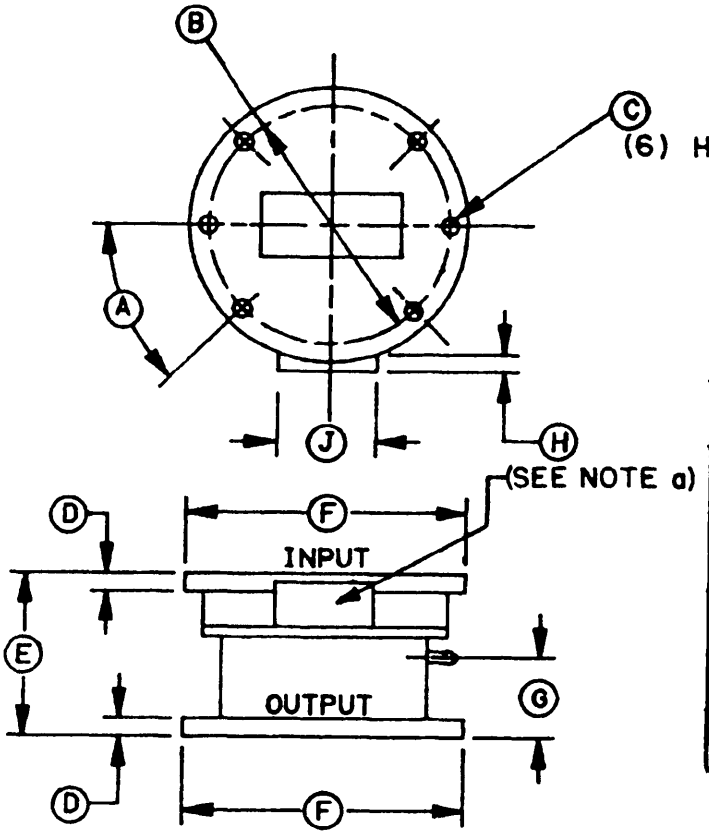
1/ See note 7

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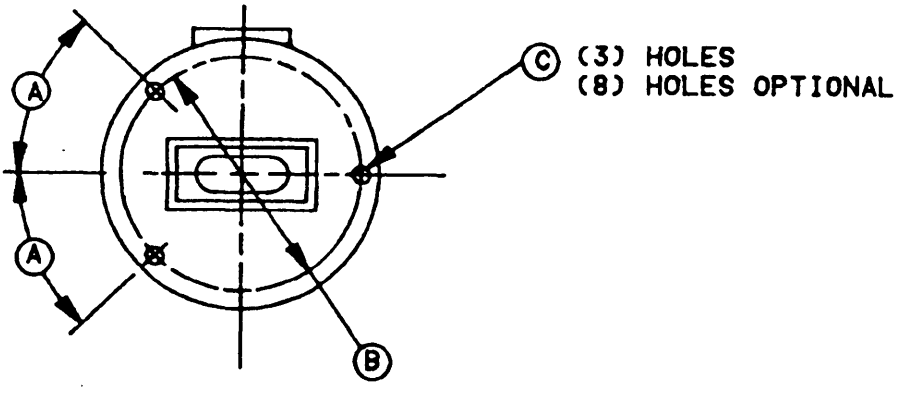
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Method	Requirement or test	Notes	Test	Conditions	Symbol	Limits		Unit
						Minimum	Maximum	
	Quality conformance inspection, part 1	1	---					
4473	Low-level VSWR	2	---					
				F1	σ	---	1.9	---
				F2	σ	---	1.4	---
				F3	σ	---	1.4	---
				F4	σ	---	1.4	---
				F5	σ	---	1.9	---
				σ' = 1.05				
4416	Insertion loss	---	---	F = F3	Li	---	0.5	dB
4452	Flat leakage power, peak (max)	3	2	Po = 4 kw±10%	Pb	---	100	mw
4452	Flat leakage power	---	2		pf	---	50	mw
4452	Spike leakage energy	---	2		Ws	---	0.10	erg
4494	Position of short	4	2		Distance	---	±0.015	Inch
4488	Arc loss	---	1		Loss	---	0.80	dB
1027	Temperature cycling (nonoperating)	---	---	1 cycle	---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	5,6	---		---	---	---	---
4471	Recovery time	---	3		t	---	5	μs
	Quality conformance inspection, part 3							
---	Life test (1)	8	3	Group D	t	4,000	---	hrs
---	Life test (2)	---	3	Group D	t	1,000	---	hrs
---	Life (1 and 2) -test end points:							
4471	Recovery time	---	3		t	---	10	μs
4452	Flat leakage power	---	2		pf	---	70	mw
4452	Spike leakage energy	---	2		Ws	---	0.1	erg
4416	Insertion loss	---	---	F = F3	Li	---	0.8	dB
1027	Temperature cycling life test	9	---	10 cycles (min); Group C	---	---	---	---

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
C	0.216 (5.49)	0.226 (5.74)
F	2.100(53.34)	2.120(53.85)
G	0.900(22.86)	- - - - - -
H	- - - - - -	0.130 (3.30)
Quality conformance inspection, part 2		
B	3.245(82.42)	3.255(82.68)
J	- - - - - -	1.500(30.10)
Periodic inspection (see note 9)		
A	44°	46°
D	0.235 (5.97)	0.265 (6.73)
F	3.610(91.69)	3.640(92.46)



NOTES:
a. The reservoir overall dimensions shall not exceed the dimensional requirements and shall be physically and functionally compatible and interchangeable in all applications to the tube types in note 7.
b. Silver plate 100msi minimum, or equivalent.

FIGURE 1. Outline drawing of electron tube type 8919.

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. A swept-frequency method may be used for this test.
3. The maximum peak flat leakage shall not exceed the specified limit when tested from zero to 4 kw peak power.
4. The position of short shall be 0.085 inch (2.16 mm) further from the magnetron within the limits specified.
5. The input flange only of the TUT shall be immersed in water during this test.
6. At the conclusion of this test, the TUT shall satisfy the specified recovery time test.
7. This type physically and functionally replaces types MD-80C17 and VDC-1029 (NSN 5960-00-106-0423), and can replace type 5865 (5960-00-262-0174). This information in no way implies that the manufacturer's types are suitable as substitutes for the military types.

   NOTE: When type 5865 is replaced by type 8919, the keep-alive (ignitor) supply shall be disabled and the lead removed or tied securely.
8. This test shall be performed only during qualification inspection.
9. This test (or figure 1 dimensions) shall be performed (or checked) during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, these dimensions shall be made as a part of quality conformance inspection, part 2, with an AQL of 6.5, inspection level S3 (code D for figure 1 dimensions). The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
10. Revision letters are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - ER
Navy - EC
Air Force - 85

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Air Force - 17, 99
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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